

Nathaniel Oliver

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Education

University of California, San Diego

M.S. Data Science, GPA: 3.925/4.0

La Jolla, CA

Expected June 2027

University of California, San Diego

B.S. Mathematics & Economics, Minor in Data Science

La Jolla, CA

June 2025

Technical Skills

Languages: Python, SQL, TypeScript, JavaScript, R

Frameworks: LangGraph/LangChain, Pydantic AI, FastAPI, React, PyTorch, PyTorch Geometric, pandas, scikit-learn, XGBoost/LightGBM, PySpark

Developer Tools: Git, GitHub, Claude Code, pytest, CI, Docker, Linux, AWS

Databases: PostgreSQL, pgvector, Supabase, FalkorDB, SQLite

Misc: knowledge graphs, GraphRAG, RAG, MCP, evals, Claude/OpenAI APIs, prompt engineering, REST APIs, statistics, ETL/ELT

Projects

ARTie: Agentic Resume Tailoring (repo) | *Python, LangGraph, FastAPI, Postgres, pgvector* 2026 – Present

- Built a **multi-agent LangGraph** pipeline that ingests resumes, LinkedIn, and GitHub into a **knowledge graph**, retrieves experience and skill facts by **GraphRAG**, and extracts user preferences from chat into stored facts under a **semantic abstraction layer**
- Drives tailoring through a **two-agent planner and tailoring loop** selecting revision strategies under an **epsilon-greedy policy**, each attempt scored by a **composite reward** (ATS score, semantic similarity, **faithfulness check** against the graph): **20% average ATS uplift** on a small sample

E-Commerce Session Analytics & Recommendation (repo) | *PyTorch Geometric, RelBench, XGBoost* 2026

- Built a **slice-aware** benchmark pitting a **heterogeneous two-tower GraphSAGE GNN** against a tuned **XGBoost LambdaRank** ranker on **218K** browsing sessions under a **budget-matched Optuna** sweep, with controlled **ablations** on receptive-field depth and scoring head
- Audited the **temporal correctness** of a custom **RelBench** graph with purpose-built reachability and subgraph-overlap probes, catching **two distinct data leaks** that had inflated earlier results, yielding the honest finding that **XGBoost wins every slice** (Novel Recall@10 **0.174 vs. 0.103**)

Experience

AI Engineer Intern

Aug. 2026 – Present

Agos

San Francisco, CA

- Building the **memory layer** behind a **voice agent**: a **claim ledger** whose retrieval scores every claim by **provenance, recency, and support**, resolves entities and expands **two hops** to surface conflicts, selects and **quotes evidence under a context budget**, exposes memory to a task as a **read-only verb**, and retrieves speculatively against a sentence **still being spoken**
- Building the **benchmark** that measures the memory layer itself: a **stratified 30-case LongMemEval** sample fixed by a **seeded manifest** with benchmark and dataset **pinned to exact revisions**, running a **baseline ladder** (full history, **BM25**, dense, hybrid) against extracted memory under an **identical reader and official judge**, scored on QA accuracy against **context size, cost, latency, and source-support failures**
- Built a **dependency-free assertion toolkit** keeping design documents, code, and **CI** from drifting apart: **pins a document sentence** so breaking a stated contract breaks a test, **ratchets measurements** against silent regression, and reads **three-dot diffs** so a branch is judged on what it changed

Data Science Intern

Jan. 2026 – Aug. 2026

IDX Exchange

Remote

- Built a **streaming pipeline** on **AWS** carrying cleaning, geospatial filtering, and **MCAR/MAR/MNAR missingness analysis** inline: **5-minute delta ingestion** into **S3**, **PySpark** on **Glue**, a **SageMaker Feature Store**, and a queue-driven **Postgres** sync, having processed **100K+** housing listings
- Built the **production model**, a tuned **XGBoost** predicting single-family California home prices at **7.74% MdAPE** on **200K+ CRMLS** listings and held across **scheduled retrains**

Economic Research Assistant

Apr. 2024 – June 2025

UCSD Department of Economics

La Jolla, CA

- Designed and ran the empirical studies behind ongoing research (**Python, R, Stata**), applying **time-series structural-break** (Bai–Perron, CUSUM/MOSUM) and **difference-in-differences** analysis, and **presented results to PhD researchers** to steer the direction of the work